

SecureVault – A Secure File Encryption and Decryption System

-Cipher Syndicate

Problem Statement: [CC-01-S3] Secure File Encryption & Decryption Tool

The Problem At Hand:

In today's digital environment, users often store sensitive files (documents, personal data, credentials etc) without proper encryption. Existing tools are either too complex, not user-friendly, or lack integrated file organization and security. This creates a need for a simple and secure solution for everyday users.

Our Proposed Solution:

Hence, we have decided to build SecureVault: a desktop application that allows users to securely store files inside password-protected folders. These files are encrypted using strong cryptographic methods and can only be accessed with the correct password.

Core Feature:

- ❖ The key feature of SecureVault is its combination of:
 - Strong encryption (AES-GCM via OpenSSL in C)
 - Simple graphical interface
 - Vault-based storage system that allows for easy organization within the app
 - ❖ This combined with its ease of setup and installation allows users to securely *encrypt, store, preview, and decrypt* files easily with minimal effort and no technical expertise.
 - ❖ Users can also keep *logs* of their activity, allowing them to track the date and time of the encryption and decryption of files.
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Target Users:

- Students managing academic or personal files.
- Individual storing confidential data
- Users who require simple and secure file protection without technical complexity